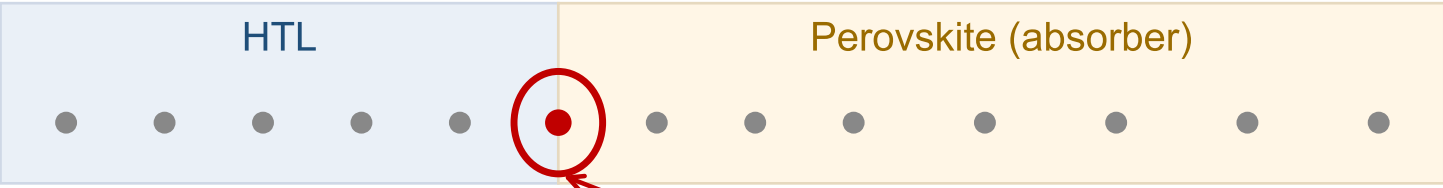


Why SolarLab meets the heterointerface Auger double-count — and SCAPS does not

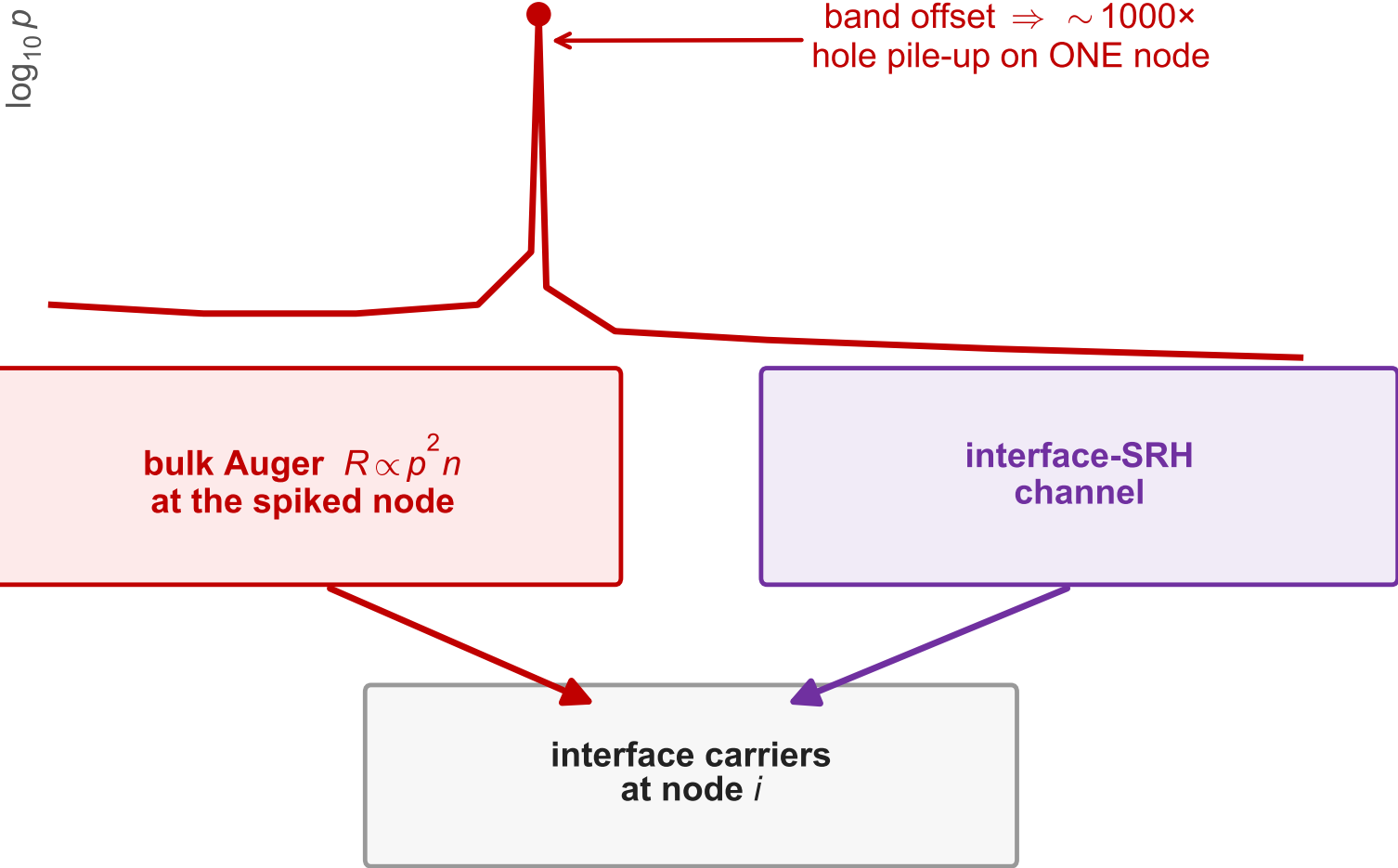
Same physics (band offset, Auger, interface traps) — different ARCHITECTURE / bookkeeping.

SolarLab — one continuous drift-diffusion grid

the junction is just another BULK node



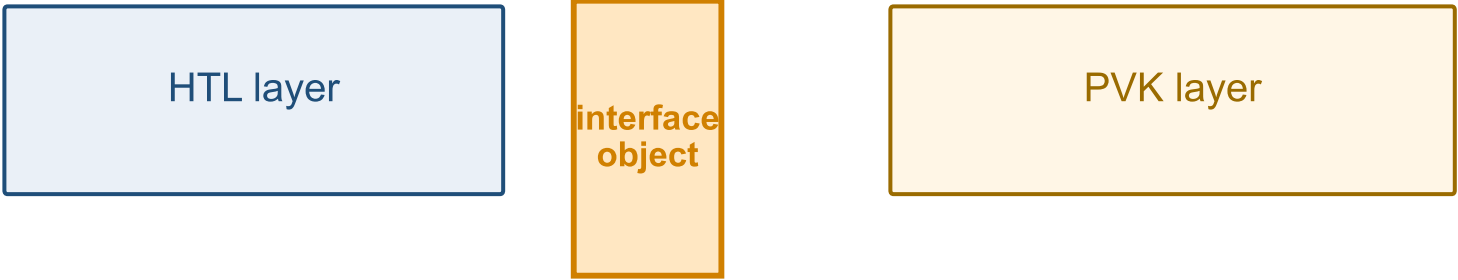
node i = junction
(a BULK node)



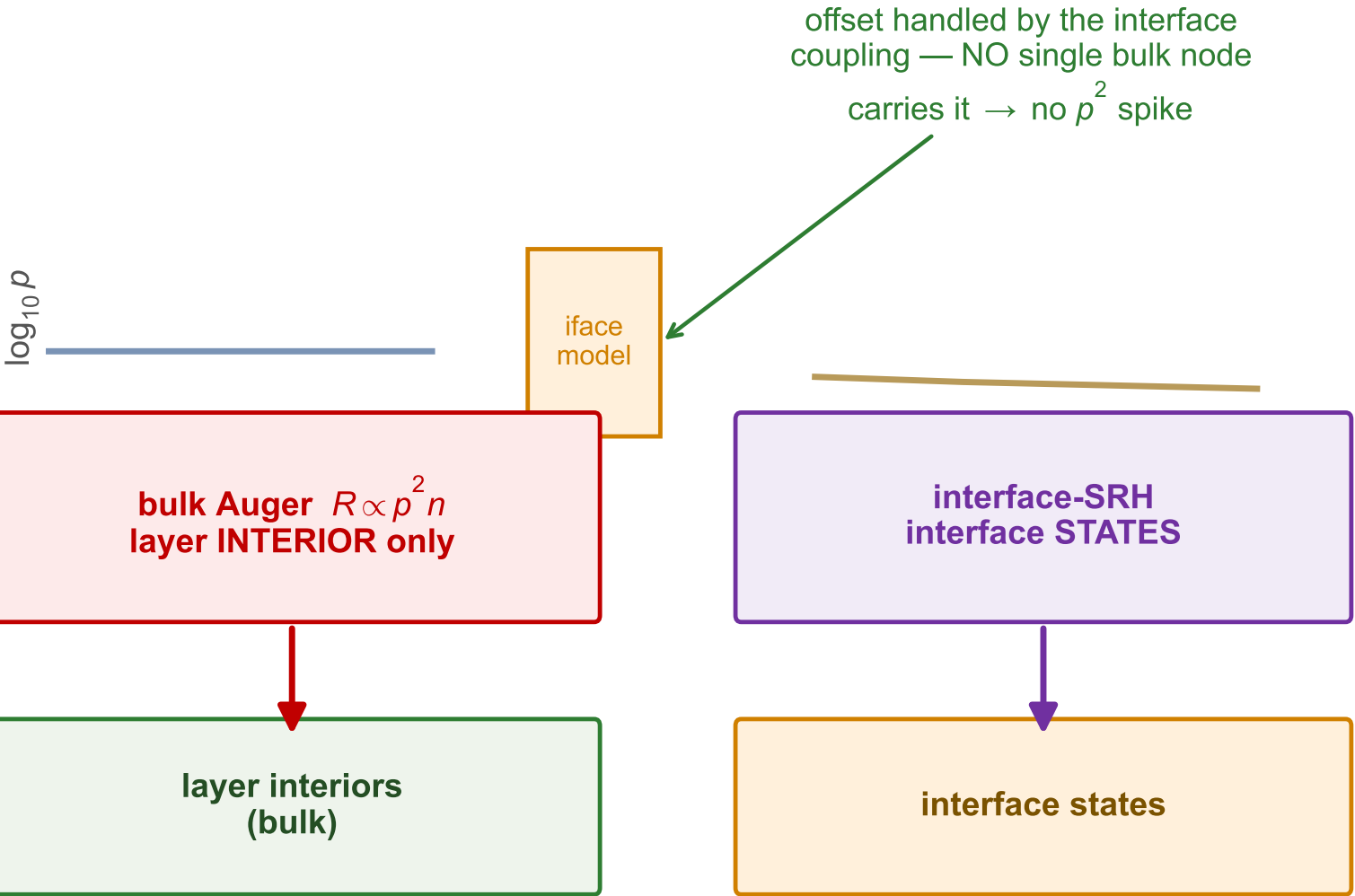
SAME interface loss counted TWICE => double-count

SCAPS-1D — layers + an explicit interface

bulk in layer interiors, interface on interface states



DISCRETE layers + an EXPLICIT interface between them



each carrier feeds ONE channel => interface counted ONCE

SolarLab's continuous grid makes the junction a bulk node — the absorber Auger formula sees the pile-up AND interface-SRH counts the same loss.

SCAPS' layer+interface split routes bulk and interface recombination to separate targets. SolarLab's proper fix = interface-plane states; the de-spike f is its cheap scalar emulation.